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Semiotics and Human-Computer Interaction

Clarisse Sieckenius de Souza

Pontifical Catholic University of Rio de Janeiro (PUC-Rio)

Rio de Janeiro, Brazil

Abstract

This chapter presents a broad-brush picture of what Semiotics is and how it can contribute to advance HCI research and practice. Content and references privilege clarity and cohesion of presentation, rather than coverage of existing work. It begins with a concrete example of user-system interaction from which the author extracts the elements to introduce some fundamental concepts in semiotic theories and show how they can be applied to HCI. The chapter also includes an overview of work done to date in this interdisciplinary field, as well as a final commentary on potential avenues for future research.

Keywords: Semiotics, computer-mediated human communication, metacommunication through digital proxies

Semiotics and Human-Computer Interaction

This chapter presents a broad-brush picture of Semiotics and how it can contribute to HCI. Content and references privilege clarity and cohesion of presentation, rather than coverage of existing work. The list of references comprises publications by authors with very different perspectives, which interested readers are invited to explore.

Although the definitions of Semiotics may vary substantially (Nöth, 1995; Trifonas, 2015), the primary aim of a semiotician's work is to investigate signs and signification (Chandler, 2002; Eco, 1979). A sign can be defined as anything that can be used to represent something else and signification is the process by which signs come into existence. This is not an exclusively human capacity. Animals, for example, signify the world around them and there is controversy about whether machines can do it, too (Nadin, 2007; Nake & Grabowski, 2001; Nöth, 2002). Since signs are not necessarily verbal, interdisciplinary researchers have taken human-computer interaction as the object of Applied Semiotics studies (Andersen, 2001; Barr, Biddle, & Noble, 2005; Benyon, 2001; de Souza, 1993; Goguen, 1999; Keeler & Denning, 1991; Mullet & Sano, 1995; Nadin, 1988a; Nadin, 1988b). A semiotic account of humans in HCI says that signs are produced by users, but not only users! Systems designers and developers, for example, engage in signification processes that are just as important (and as deserving of rigorous investigation) as the users' (Andersen, 1992; de Souza, 2005a; Kammersgaard, 1988; Nadin, 2001), and so do other stakeholders in the Information and Communication Technology arena.

A Semiotic Vignette

The following HCI vignette shows what a semiotic perspective looks like, with computer-generated and computer-mediated signs unfolding in a piece of fiction based on true factsⁱ.

Context

Mr. User is a retired engineer who spends much of his time on the Internet, communicating with family members and friends, watching videos, reading the news and browsing for shopping opportunities. He just discovered that his favorite browser has an Autofill tool that can save him time and memory. He decides to explore Autofill and begins by reading online help information, which says:

“Save time with Autofill. It stores the information you type in when filling out a form and allows you to reuse it the next time you need to inform your credit card number, name and address. You can add, update, and delete information whenever you wish. Autofill is online registering and purchasing made easy!”

Encouraged by this message, Mr. User engages in interaction aiming to configure the use of Autofill. The 6-step storyboard in Figures 1, 2 and 3 shows how his experience develops.

The annotations on screen sketches indicate the sequence and the logic of interaction.

Storyboard

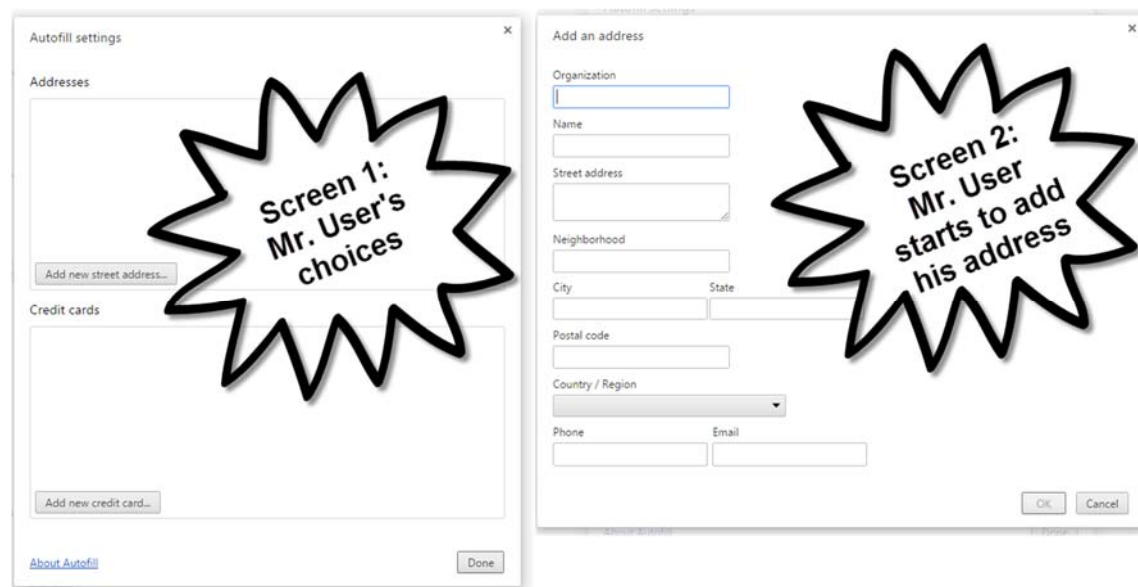


Figure 1: Storyboard steps 1 and 2

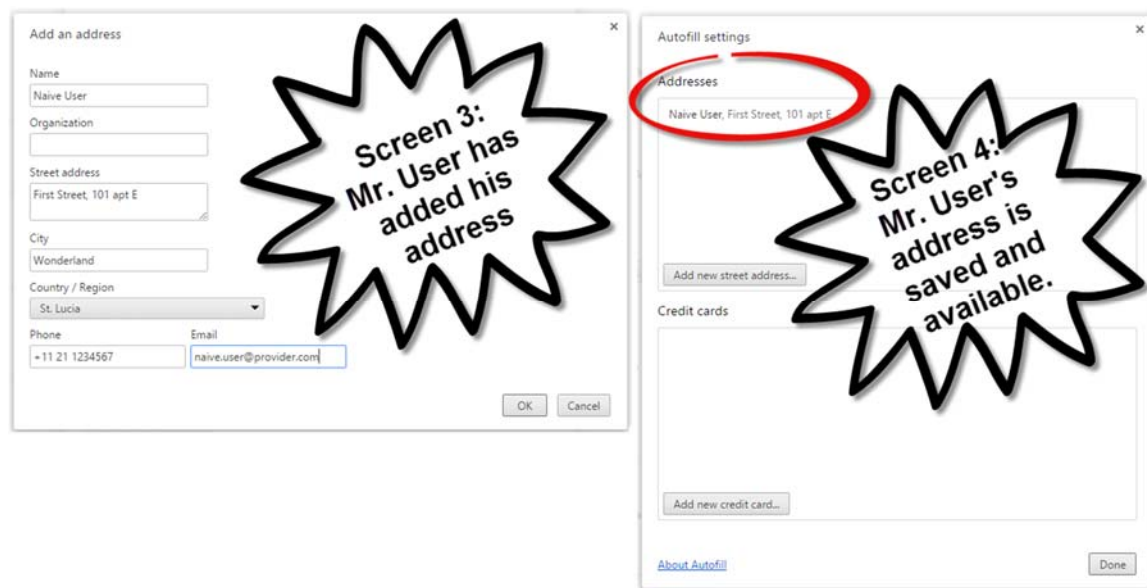


Figure 2: Storyboard steps 3 and 4

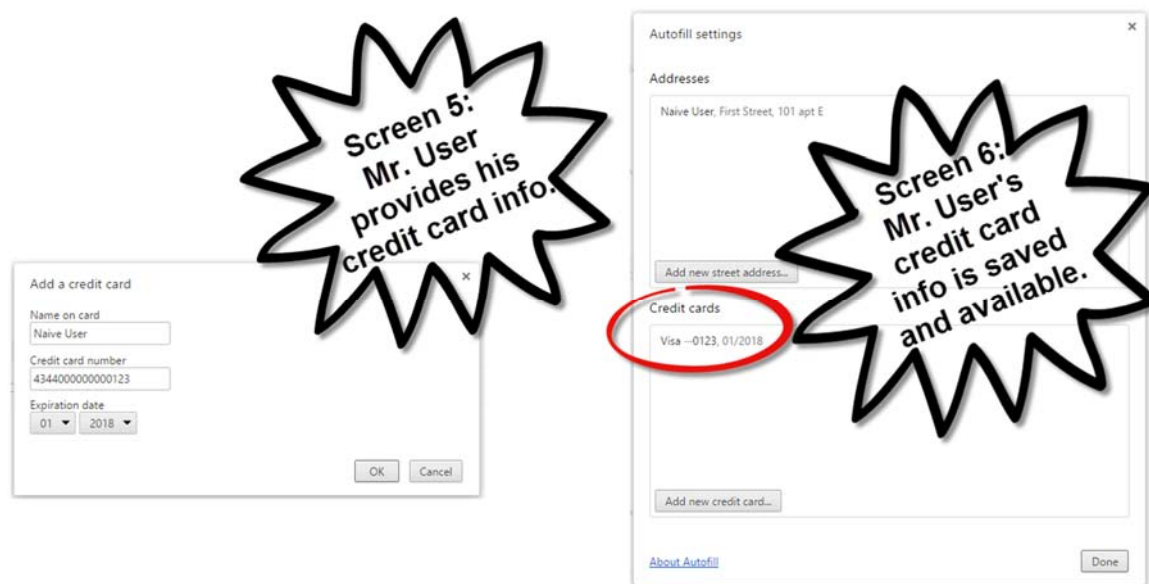


Figure 3: Storyboard steps 5 and 6

The vignette illustrates how a naïve user can decide to store critically sensitive information in his local browser's database without ever questioning whether this is a risky decision. To be sure, in 2015, when Autofill was available, online theft of credit card information was a nightmare for credit card administrators and many clients. Mr. User's behavior is not surprising, however, and Semiotics can help us appreciate why, in greater depth and detail.

HCI as Computer-Mediated Human Communication

Except for the content of annotations, are the signs appearing in the storyboard's screens: (a) the browser's; (b) the user's; (c) both the browser's and the user's; or (d) neither the browser's, nor the user's? Probably, (c) will be the most frequent answer. On Screen 5 (Figure 3), for example, labels like “*Name on card*” and “*OK*” have been generated by the browser, whereas “*Naïve User*” has been typed in by the user.

Effects of communication: Acting through interface signs

The expression ‘Name on card’ is produced by a program, which is apparently the agent of the signification process required to communicate something to users through the system's interface. But what does it communicate? Since the meaning of interface signs can be elicited by verbalizations ([Andersen, 2001](#)), in Figure 4 the balloons show how subsequent interaction steps might probably be verbalized by users.

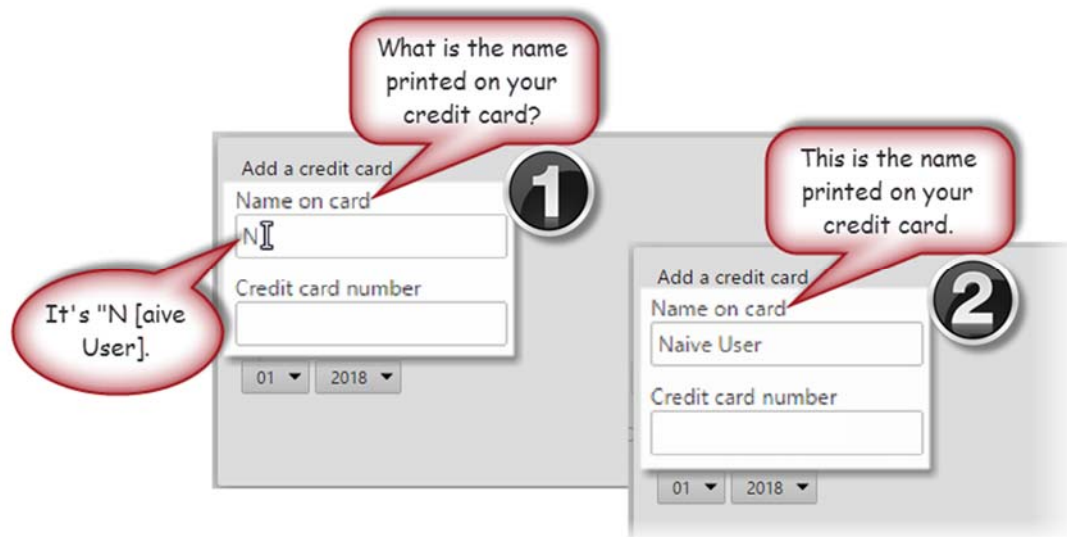


Figure 4: Verbalizing interface signs

On the left-hand side of Figure 4, the label “Name on card” is verbalized as a request (see “1”), to which Mr. User reacts by typing the name that appears on his credit card. His reaction is verbalized as: “It’s N...” (the snapshot is taken before Mr. User has finished typing his entire

name). On the right-hand side of Figure 4, “Name on card” is verbalized differently, as an assertion rather than a request (see “2”). Just like language used in communication, which can change the world and the mental state of speakers and listeners, interactive requests and assertions in HCI are also speech acts (Austin, 1962; Searle, 1969), achieved through computers.

A user’s expectations, beliefs, technical knowledge and trust in developers may lead to completely different verbalizations. Starting with simpler variations, someone might justifiably verbalize “Name on card”, in all turns of interaction shown in Figure 4, as a single assertion: “This is the name printed on your credit card”. It expresses a *false* statement before the user types the appropriate name and a *true* one afterwards. In this case, as a reaction to an indirect speech act (Searle, 1975), the user corrects a wrong system state. In the previous one (Figure 4), the user responds to a system’s request. Although these are different interpretations, as far as the system is concerned, both enable the system to move to the same expected step successfully.

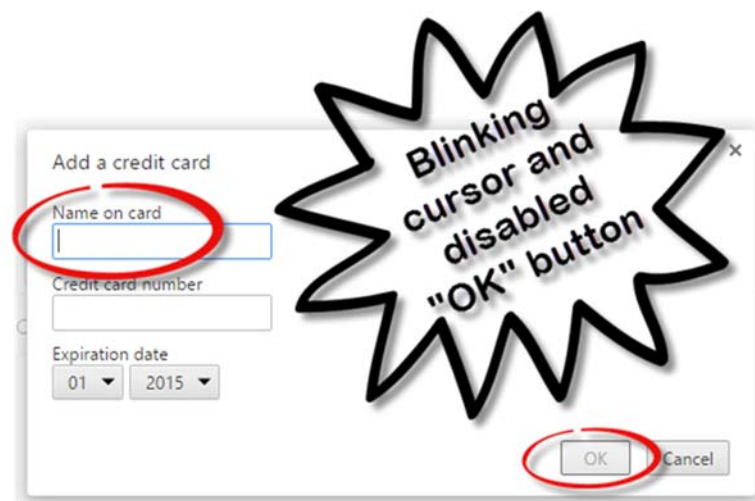


Figure 5: Evidence for interpreting “Name on Card” as a request

In Figure 5 we take a closer look at how speech acts take effect in HCI and see why one interpretation is in fact more productive than the other. When Mr. User opens the dialog to add a new credit card, two non-verbal interface signs communicate something that is not strictly

consistent with the truth condition interpretation. The cursor is positioned inside the “Name on card” text field and blinking (the text field’s border itself has a different color than everything else on screen). Besides, the “OK” button is disabled, preventing the user from storing credit card information if the value of this particular field is not changed. An adequate interpretation of these signs has little to do with truth conditions and much to do with requests. Moreover, since it is impossible to submit a form where the card holder’s name is missing, this may be justifiably taken to mean (or signify) that the system *knows* what counts as a valid form. This conclusion would, however, be contradicted in subsequent steps. The system used in this chapter’s vignette allows the user to store invalid information, like an expiration date prior to the current date, or an empty credit card number field (which would seem to defeat the purpose of Autofill). In view of such behavior, the user might draw a final unfavorable conclusion, that Autofill cannot protect users against errors and slips.

A deeper analysis of signs and signification shows that the user might once again have reached the wrong conclusion. A savvier user, well acquainted with the risks of storing credit card information in a browser’s database, can use automatic form filling in clever ways. For example, she can fill up the form with ‘quasi valid’ or ‘quasi complete’ information. In this case, she might intentionally void the credit card number field, while filling up the card holder’s name and the expiration date correctly. The effect of this intentionally incomplete response to the system’s request is that risks entailed by information theft are substantially decreased. The picture is completely changed and Autofill now figures as a cleverly designed component, rather than a sloppy piece of software. Which one is the case?

Regardless of the answer, by going beyond the meaning of interface signs (a semantic perspective) and into the effects that they are meant to achieve (a pragmatic perspective), we

gain useful insights for the design of human communication through computer proxies. For example, in Figure 6 the sketched interface explicitly communicates that Autofill designers intentionally allow for partial information storage, so that users can choose the level of risk they want to take.

Learn more.'. Below this is a required text input field for 'Name on credit card: (required)'. A section titled 'Optional Fill-In' contains a 'Credit card number:' input field with a note 'Leave this blank if you want more security and flexibility.' and a 'Learn more.' link. Below the number field is an 'Expiration date:' section with 'mm' and 'yyyy' dropdown menus. At the bottom are links for 'Help & Documentation', 'OK', and 'Cancel' buttons."/>

Figure 6: A redesigned interface for the Autofill dialog

Who is engaged in communication

By seeing themselves as speaking to the users through the system's interface, interaction designers can explore more often and deeply their own pragmatic awareness of offline communication experiences. Pragmatics can help them anticipate numerous breakdowns that have to do with shared social practices, especially if they add to their capacity of looking sharply at design *objects* that of looking at design *subjects*, which include themselves.

All semiotically informed accounts of HCI view human-computer interaction as a special case of computer-mediated human communication (Andersen, 1997; de Souza, 2005b; Nadin, 1988a). Although users are and should be at the center of HCI concerns, this does not come at the expense of all other parties whose messages and intent are expressed during interaction.

Designers and developers are among these, but there are more players in the game like other users, systems owners, commercial companies, public and private organizations, and many more. All speak through technology deployed to benefit their own interest and the interest of those they care for.

To illustrate the involvement of different stakeholders, we proceed with Mr. User's story, now supposing that the company that developed and distributes his browser requests access to the users' social network. Mr. User grants them the requested permission and the company uses the fact that Mr. User has turned on his browser's Autofill function to persuade his friends to do the same.

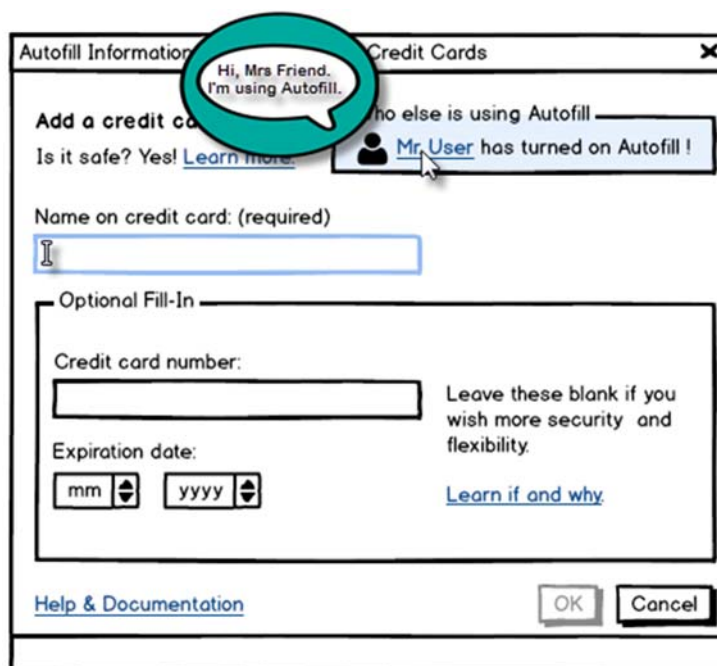


Figure 7: Using social network information from a user's account

Figure 7 presents a sketch of what the interface might look like for Mrs. Friend, who is part of Mr. User's social network. The text message inside the shaded rectangle on the upper right side of the screen is a system's message telling Mrs. Friend that Mr. User has turned on Autofill. However, when she hovers the mouse over the link with his name, a balloon message

pops up saying: “Hi, Mrs. Friend. I’m using Autofill.” In spite of the first person in the message, is it really Mr. User who is speaking? This communication is sent on Mr. User’s behalf by the system, which in turn speaks for the company. He is most probably not aware of it, and therefore does not intend it to be sent.

Systems do not have intentions of their own and there is no communication without intentionality (at least presumed intentionality). Therefore, in order to achieve human-computer interaction, systems designers and developers must impart intentionality to systems. Although many decisions about the communicative intentions are derived from impersonal task and domain models, the collection of examples in this chapter shows that decisions can also derive from personal intuitions, values and cultural practices that designers consciously or unconsciously hold. These eventually get inscribed into the systems they design (de Souza, Cerqueira, Afonso, Brandão, & Ferreira, 2016). The contribution of a semiotic perspective is to provide conceptual models, methods and tools to support intentional design of implicit and explicit computer-mediated human communication through systems interfaces (Brejcha, 2014; Mullet & Sano, 1995; Pereira & Baranauskas, 2015).

Communicating with the mediation of an automaton

Computers play the role of medium and also that of proxies speaking on behalf of people to other people (de Souza, 2005a; de Souza, 2005b; Fogg, 2003; Light, 2004; Nadin, 2001; Nake & Grabowski, 2001). In the latter case, a computer system executes one or more programs that contain all and only the rules that determine the proxy’s capacity to mediate communication between the human parties involved in it. These programs can be interpreted as implementations of human communication models, through verbal and non-verbal signs (Andersen, 2001; Brennan, 1990; O’Neill, Benyon, & Turner, 2002). One of the hard challenges for HCI design is

that computer programs *are* automata, everything they do is strictly determined by rules governing their response in anticipated situations that may or may not occur. This does not happen to (and is therefore not expected from) humans, whose social behavior is hardly ever the same, changing and adapting constantly to the most minuscule difference in familiar situations. The fact that humans' computer proxies are governed by specified rules, even when learning from their environment, means that their behavior is, at least in theory and unlike our, predictable (and hence *replicable*), over time and space. This opens new possibilities for social interactions whose nature and effects are, as of yet, relatively unknown (Gaines, 2015; O'Halloran, 2015). As a thought exercise, let us project some future scenario where most users will create several software proxies for themselves. The programs governing interactions with others (possibly others' proxies) will strongly signify their creators' beliefs about an important part of what they *are*, a powerful sign for investigating human *subjectivity* (Tanaka-Ishii, 2015; Turkle, 2005), as well as for investigating human *sociability* and *culture* (Brejcha, 2014; Brejcha, 2015; Pereira & Baranauskas, 2015; Salgado, Leitão, & de Souza, 2013).

The popularization of social communication through computer proxies can be the result of many efforts being carried out in the field of End User Development (Lieberman, Paternò, & Wulf, 2006). A larger number of users will be able to write programs, usually called scripts, to do various things for them. One thing that they can already do is to send birthday messages to friends whose birthdays are marked on their calendar. This sort of communication has the potential to outlive the senders and receivers of such messages and create a new object of study for post-mortem interaction research (Pereira & Maciel, 2013).

Practical evidence of some of the consequences of our computational *persona* reaching space and time extending beyond the physical limits of our existence comes from Google. In

2013, they launched the Inactive Account Manager (IAM), whose purpose is to help users *plan their digital afterlife* (Tuerk, 2013). Users to tell Google what to do with their account information and services if they die or are no longer available. For example, an active user may have a shared agenda that automatically sends out meeting invitations to all team members of a project, on every fourth Friday of the month. If this user dies unexpectedly, this person's colleagues will keep receiving his or her invitations for monthly meetings for as long as the account is active. With IAM this can be prevented since Google users can, among other options, designate trusted individuals to take the necessary measures under specific circumstances.

The birthday card and IAM examples can be combined in script scenarios using IFTTT ('If This Then That'), an easy-to-use cloud-based tool that enables users to program events using apps and web services. One popular 'recipe', as IFTTT scripts are called, sends one's friends an automatic Happy Birthday messageⁱⁱ. By simply connecting Google Calendar and Facebook channels, a user can project his social interaction to time and space beyond the limits of physical life. No programming skills are required; all the users have to do is to 'point and click', while IFTTT guides them through a few steps in the creation of an actual running instance of the recipe. This is already an open door to posthumous interactions, which can turn into whimsical afterlife conversations if, while living, recipients of birthday messages write their recipe to send back 'thank you' notes to their friends.

Semiotic theories and approaches

Signs and signification processes mediated or automated by computer programs can be explored using different semiotic theories. This section presents a brief overview of concepts used in semiotically informed HCI research and illustrates research done to-date in this area. Examples are based on the same scenarios used in previous sections.

Semiotics

Semiotics has a long history in Philosophy (Eco, 1986). In spite of many schools and theories (Chandler, 2002; Nöth, 1995; Trifonas, 2015), most contemporary work in Semiotics stems from two traditions, established by Ferdinand de Saussure (1995), in Europe, and Charles Sanders Peirce (1992; 1998), in North America.

De Saussure focused on Linguistics and the role of signs in social life. *Semiology*, as he coined the discipline, should investigate not only the nature of signs and the laws governing the conceptual structure and meaning of language, as an abstract system, but also the situated use of signs and sign systems by language speakers, in psycho-socially determined real-world conditions (de Saussure, 1995). Peirce focused on Logic, Epistemology and the Philosophy of Science, trying to establish the conditions under which one signifies reality. *Semiotic*, as he coined the discipline, should investigate the nature of signs and sign processes through which minds continually construct and evolve the meaning of whatever becomes the object of observation, experience, reasoning or imagination (Peirce, 1992; Peirce, 1998).

De Saussure and Peirce had different definitions for their object of study. The Saussurean sign is defined as a unit formed by a concept (*signified*) and a sound pattern (*signifier*). The connections between signified and signifier are established systemically by paradigmatic and syntagmatic relations. Paradigmatic relations spring from distinctions that, in some specific dimension, define classes of signs that are interchangeable (*i. e.* belong to the same paradigm) or not interchangeable (*i. e.* belong to distinct paradigms). Syntagmatic relations express the compositional principles behind complex signs, which are combinations of atomic or other complex signs from different paradigms. This was the basis for a greatly influential school in

Linguistics and other disciplines called structuralism, whose impact has even reached the study of formal languages and computing (Chomsky, 1959; Hopcroft, Motwani & Ullman, 2013).

De Saussure's interest in how signs occur in real psycho-social situations has also led to fundamental progress in the study of natural languages. Unlike in abstractly defined conceptual systems, in the real world variability is everywhere, some of which is predictable and acceptable, while other leads to breakdowns and disruption. *Langue* and *parole* are the two Saussurean concepts used to distinguish between the abstract conceptual dimension and the situated psycho-social dimension of language, respectively. They opened the door to the study of formal linguistics, on the one hand, as well as of various other fields of language studies focusing mainly on Psychology, Sociology and Culture.

De Saussure emphasizes that the fundamental theoretical relation between signifier and signified is arbitrary. There is nothing in the latter that determines the shape of the former, although such determination may be occasionally verified, as in onomatopoeic words like "click". In HCI, however, although computer proxies' signs are arbitrary with respect to the form that they take in a system's interface, they are causally related to the rules that govern the system's behavior while in use. Therefore, not only the theoretical arbitrariness of the sign, but also the difference between *langue* and *parole* are blurred in a Saussurean analysis of interactive systems. A programmer's personal use of language (*parole*) becomes the system's rule (*langue*) once it is encoded in software.

Peirce, unlike de Saussure, defined the sign as a triad composed of a representation (the *representamen*), an object and an interpretation (the *interpretant*). Interpretation is the necessary link that binds an object to some representation and thus creates meaning. The binding can be due to habitual associations governed by personal experience or social conventions, as well as to

sudden realizations or intuitions about previously unsuspected relations. In Peircean Semiotics, meaning is the result of interpretive mediation that takes something – actually, anything taken as a contingent representation – to stand for something else (its object). Unless this mediation is in place, no sign can exist. Peirce’s *interpretant* explains, for example, why the English word “cancel” (the *representamen*) stands for some action in the world (the object). A collective of minds establishes a conventional connection between the linguistic representation and the phenomenon or event in reference. This object can also be signified by non-linguistic representations. The image of an interface control like  or , as well as the deletion of an item on a web page list, both can be taken to *signify* the act of canceling, for somebody, in some respect or capacity. Explanations why representations signify cancelations, before or after the fact of signification, are an essential for a Peircean analysis, which may involve shorter or longer chains of meaning associations, with implicit and explicit inferences.

A special feature of the mediating interpretation is that the mediator constitutes, in itself, another sign. This is to say that, among other things, higher-order interpretations may occasionally bind a previous interpretation to its object if some mind performs the conscious or unconscious mental operation required for it. For example, new (higher-order) interpretations (meanings) of *cancel* signs can be continually generated by computer users over time. This process, called semiosis, can take indefinitely many shapes, depending on each user’s experience with systems. We may encounter surprising elements in semiosis, like the label “cancel”, for instance, leading to meaningful mental images of an online purchase cancelation, which in turn lead to meaningful mental images of why this cancelation was made (for example, exasperating delays in delivering the purchased goods), which in turn lead to meaningful mental images of not trusting certain online businesses, and so on. Hence, “cancel” may signify flexibility and trust for

some, while signifying nothing of the sort for others. Although theoretically unlimited in its shape and duration, semiosis is halted and resumed for practical reasons related to physical, psychological and cultural determinations upon human experience.

To illustrate the different perspectives on HCI gained with these two theories we go back to the vignette presented in the previous section. As already mentioned, the Saussurean concepts of *langue*, *parole* and the arbitrariness of the sign can be traced in the implementation of the interface language through which Mr. User communicates with Autofill. Screen layouts shown in the storyboard suggest certain paradigmatic and syntagmatic relations. For instance, whereas buttons and links are paradigmatically related to each other as triggers of major state changes in the interface, buttons and text areas are not. Filling up text fields typically causes more local (minor) changes of states in the interface compared to pressing a button to submit a form. Syntagmatically, the visual structure of the interface is such that the “OK” button, for instance, occupies the last *reading* position for Western users. Other interactive controls will most probably be scanned by their eyes *before* they reach the “OK” button.

Switching to a Peircean perspective, by means of mental interpretation processes, Mr. User, who is a Western user, may interpret the first Autofill dialog window presented in the vignette’s storyboard as a representation of an automated process that facilitates his interaction with online forms. In this way, not only does he establish a productive signifying relation between the representation he sees on screen and the encoded behavior of the system, but he also imparts a positive meaning to the interaction itself by associating the outcome of interaction with desirable future scenarios. Compared with a Saussurean account, which at first pass gives us more insights about syntax and form, a Peircean account guides us more directly into certain meaning dimensions related to the user’s personal experience, expectations, beliefs, knowledge,

perceptions, and so on. Moreover, the ongoing process of semiosis can elegantly account for changes in a user's interpretation of interaction signs. Thus, for example, if – hypothetically – Mr. User hears about criminal theft of credit card information carried out by scripts that intercept automatic form-filling functions executed with Autofill, his perception and interpretation will certainly change. Mr. User will probably be worried that he may fall victim to this sort of theft and the previous meaning he assigned to Autofill will now be a sign of innocence and credulity. Note that if it that information theft risks can be minimized or maximized depending on how the user sets the browser's security parameters, the presence of interface signs referring to security and theft prevention become crucially important to orient the user's semiosis along the desired paths. The redesign of the Autofill interface proposed in Figure 6 (see “Effects of communication: Acting through interface signs”, above) is thus an example of how increased semiotic awareness can improve HCI design.

Peircean theory includes two other elements that, in specific respects discussed below, can bring interesting insights to interaction designers. One is a particular kind of logic reasoning called abduction, also known as hypothetical reasoning. Very briefly, abduction is an inference leading to the best explanation for a temporarily intriguing fact (Peirce, 1998). Intriguing facts have unknown meanings. According to Peirce, interpretation and abduction share the same basic principles. To illustrate how the process works, let us go back to Mrs. Friend, the main character of a previous scenario in this chapter. Suppose that since she started using Facebook, two years ago, Mrs. Friend has received a birthday message from Mr. User through this system. Both messages made explicit reference to her little granddaughters, which Mrs. Friend thinks is especially nice of Mr. User to mention. Now, shortly before her next birthday, Mrs. Friend's son has a new baby boy and Mrs. Friend shares the news and dozens of pictures with all of her

friends on Facebook, including Mr. User. He congratulates her profusely, but on her birthday Mrs. Friend receives a card from Mr. User wishing her a great celebration with ‘her lovely granddaughters’.

What could this card mean? That Mr. User has inexplicably forgotten that she has a new grandson? That he has carelessly copied and pasted the text from last year’s birthday message? Both alternatives, and many others, would be potential explanations for Mr. User’s intriguing behavior. Finding the meaning of his message involves the generation and test of different hypotheses. For example, Mrs. Friend will discard the first alternative explanation because she has concrete evidence that Mr. User hasn’t forgotten that she has a grandson. He posted a comment on one of the baby’s photos just the day before her birthday. Although she does not know it, the other alternative explanation is also false. When he joined Facebook a couple of years before her, Mr. User created a script to send his friends a birthday card automatically. As a result, Mrs. Friend may conclude that Mr. User has been reusing old birthday cards and that his messages are much less *personal* than they first sounded. The conclusion may become a conviction when she hears a common friend comment that he doesn’t like getting always the same birthday messages from Mr. User (first piece of evidence reinforcing the ‘copy and paste’ hypothesis) and when, fortuitously, Mrs. Friend’s son comments that just the other day Mr. User called to ask him why he could not copy and paste text from a PDF file that he had downloaded (second piece of evidence reinforcing the ‘copy and paste’ hypothesis). The effect of such meaningful but wrong inferences is that Mrs. Friend will be very disappointed with Mr. User, and what was once the source of positive feelings is now the source of negative ones. We can wonder what might happen if Mrs. Friend ever heard about IFTTT recipes, which points to the potential value of Peircean semiotic theories in the elaboration design scenarios. They can

contribute to increase the number and diversify the sort of envisaged situations to which computer proxies of human beings (in particular an HCI designer's own) must respond computationally.

The other additional element of Peircean semiotics that is worth bringing up in this section is the role of mental habits in shaping continually evolving semiotic paths. Since humans have finite resources (mental, physical and other), although unlimited in the long run, from an individual's perspective semiosis cannot be infinite. The search for confirming evidence to support meaning generation and the generation of higher-order meaning based on previous interpretations are halted when the interpreter runs out of resources like time, information or even motivation and patience. Mental habits play an important part in the generation of hypotheses regarding the meaning of surprising facts. Once the presence of reinforcing evidence for a hypothetical meaning saturates the need for confirmation, not only does the hypothesis gain a new status in the pool of our beliefs, but it also begins to affect the way how we interpret reality. For example, it may blind us to the observation of disconfirming evidence for certain beliefs (Magnani, 2013).

The study of signification habits and how they affect interpretation has been one of Peirce's most significant contributions to Philosophy. He is the founder of Pragmatism and the precursor of linguistic Pragmatics (Nöth, 2011; Peirce, 1998), which we can define for the sake of simplicity as a current of thought or field of study that conceives of meaning as encompassing all the practical consequences of interpretation. For HCI, in particular, a pragmatic perspective can be extremely insightful because some of the thorniest issues in HCI design lie precisely in the practical consequences of encoding a particular mode and range of interpretations for the signs that digital proxies are able to handle in communication. For example, the practical

consequences of not providing any signs of trustworthiness in the design of interaction with Autofill may be that the average user generates and sustains negative beliefs about the intentions of Autofill's developers. They, in turn, with the habit of being trusted for pieces of software that they have produced, may well forget the need of reassuring and convincing their users that this component can be used safely.

In spite of the insights that abduction and semiosis can bring to how culture, context and time affect users' interpretations of interface signs, the most popular use of Peircean Semiotics in HCI has been his classification of signs into icons, indices and symbols. Graphic interface designers have been using this classification since the early 1990's (Marcus, 1992; Mullet & Sano, 1995) to build interface sign catalogs and styles. In HCI, a much simpler version of Peirce's definition is used: icons are signs whose representations resemble the object to which they refer; indices are signs whose representations are contiguously (often causally) related to their object; and finally symbols are representations that refer to their object by means of established conventions. Storyboard screens in Figure 1, 2 and 3 show examples of the three classes of signs. A push button on a graphical interface is an *icon* of a physical push button. The labels used next to textual form fields contain words that are classified as *symbols*. Finally, the automatically updated status of the "OK" button in the form, switching from enabled to disabled and vice-versa, is an *index* of the system's computation upon the content of required fields.

Precursors, Pioneers and Developers of the Field

Most of the existing work on Semiotics and HCI stems from the pioneering work of Andersen (1992; 1997) and Nadin (1988a; 1988b). Andersen has been strongly influenced by European structuralism in Linguistics, whereas Nadin is a Peircean semiotician. Bridging between de Saussure and Peirce, Eco (1979; 1984; 1986), who also investigated the semiotics of

communication, has inspired other early work in HCI, such as Benyon's (1998; 2001) and de Souza's (1993). A considerable subset of pioneers' and their followers' research was additionally influenced by Winograd and Flores's (1986) seminal views on the design of computer-supported group coordination and communication systems, known as the Language-Action Perspective (LAP). Part of it has been developed by Information Systems researchers, where the work of Ronald Stamper (Stamper, Liu, Hafkamp, & Ades, 2000) and Kecheng Liu (2000) stands out as a reference for Organizational Semiotics. Another community of researchers look at information systems as *action systems* and have been consistently exploring the consequences and possibilities of LAP. For an overview, see Aakhus, Ågerfalk, Lyytinen, & Te'eni (2014).

LAP has extended its influence in other directions, too. De Souza's Semiotic Engineering (2005b) is considerably different from most of her predecessors' in the sense that it is not an application of semiotic theories and methods to a new object of study, like in Andersen (1997) and Nadin (1988a) for example. It is rather the construction of a dedicated semiotic theory aiming to account solely for a single specific phenomenon: human-computer interaction. Along with other Brazilian researchers, de Souza defined Semiotic Engineering's object of investigation as the process of metacommunication. It is achieved by systems interfaces when activated by users for whichever purpose or reason. Metacommunication comes from the fact that systems interfaces are messages sent from systems designers and developers to systems users (top-level communication). They tell the users how, when, where and why to exchange direct messages with the system (lower-level communication) in order to achieve a certain range of effects that are compatible with the system designers' vision. In Semiotic Engineering, systems have been characterized as the *designers' deputy*, another term to denote the concept of a human proxy used in this chapter. Semiotic Engineering has also proposed its own methods for technical and

scientific studies (de Souza & Leitão, 2009), all of them centered on *communicability*, the main quality of metacommunication according to the theory.

Over the years, Semiotics has been consistently attractive for researchers interested in the analysis, the design and evaluation of human-computer interaction. The variety of approaches can be illustrated by the work of Baranauskas and Bonacin (2008), O'Neill (2008), Valtolina, Barricelli, and Dittrich (2012), Brejcha and Marcus (2013), as well as Islam and Bowman (2016). Moreover, a number of researchers working with culture in HCI have adopted Semiotics as a theoretical foundation for their studies (Brejcha, 2015; Khaled, Barr, Fischer, Noble, & Biddle, 2006; Pereira & Baranauskas, 2015; Salgado et al., 2013). This is a particularly promising area of application for semiotic theories, given the cultural determination of signification systems and the existing body of knowledge produced by semioticians themselves when discussing culture (Eco, 1979; Danesi & Perron, 1999; Sedda, 2015). Likewise, studies on creativity and entertainment, as well as on innovative user experiences have been and will probably continue to be influenced by Semiotics (Barr, Noble, & Biddle, 2007; Danylak & Edmonds, 2005; Nake, 2005; O'Neill, 2008).

The variety and depth of semiotic research in HCI shows that Semiotics can lead us beyond decisions of which images or words should be used in interface design. By framing old and new research questions in considerably different ways, it can bring innovative perspectives and technological developments to all sorts of computer-mediated human communication.

Concluding Remarks

The views exposed in this chapter underline the role of all humans involved in software development and specifically illustrate how the meanings encoded by designers and developers on the systems that they create can reach far in affecting not only individual users, but society

and culture. In this light, earlier work about contrasting the views of computers as the source or computers as medium (Reeves & Nass, 1996; Sundar & Nass, 2000) meets new alternatives. Semiotic approaches can reconcile the two opposing views and even meet the work of BJ Fogg (2003) on computers as persuasive technologies. However, there is yet much more to learn and do. One of the most intriguing directions in which semiotically-inspired research can go is foreshadowed by studies about digital legacy (Maciel & Pereira, 2013; Prates, Rosson, & de Souza, 2015a; Prates, Rosson, & de Souza, 2015b). When coupled with progress made in the area of End User Development (Lieberman et al., 2006), it brings about thought-provoking scenarios with technology outliving users (and acting on their behalf) for an indefinitely long period of time. These can in turn meet theoretical work in Semiotics about computers as semiotic machines (Nöth, 2002; Nadin, 2007), providing solid grounds for good interdisciplinary research.

Further semiotic investigations of computer proxies for human beings can also contribute to other areas of study such as Psychology, Sociology, Communication Studies, Pragmatics, and possibly others, in addition to Computer Science itself. Once we begin to probe the meaning of programs from a semiotic point of view, programs become patterned signs, whose meanings can be associated with programming languages, paradigms and styles. Programming thus becomes part of a broader cultural production process (Andersen, 1997; de Souza et al., 2016; Floyd, 1992; Goguen, 1999; Khaled, Barr, Noble, & Biddle, 2004; Noble & Biddle, 2002; Tanaka-Ihii, 2010), which can even contribute to redefine the positioning of Computer Science relatively to other disciplines and fields. The complexity and the extent of technology's impact in contemporary life calls for theoretical foundations and perspectives that can bring together a wide variety of topics that, so far, have been studied in isolation. Andersen (2001) believes that

“Semiotics is the Mathematics of the Humanities”, an overarching theory that can have a powerful unifying effect in HCI as well as in other areas of Computer Science and Information Technology. It is thus worth mentioning, in closure to this chapter, that in his book entitled *Living with Complexity*, Norman (2010) has an entire chapter dedicated to Semiotics. This may be an important sign of how different traditions in HCI studies can begin to talk to each other and bring up exciting new insights to the field.

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ⁱ All interactions have been functionally and interactively tested by the author in November 2015 with existing systems whose identity is not provided in order to keep the focus of narrative on technology, rather than companies.

ⁱⁱ <https://ifttt.com/recipes/114870-wish-your-friends-a-happy-birthday-on-facebook>